

V2 · RMSE(S_t) · lookback 1 hour · rolling minutely

Stock-path RMSE: Monte Carlo mean path (1,000 paths) vs historical close. Continuous RTH (overnight / weekend gaps removed).
7-day = 9-15 Mar 2023 (1,945 minutely calibrations). 1-day = 15 Mar 2023 (389 minutely calibrations).

SPY — RMSE(S_t)

Model	7-day	1-day	best window
GBM	1.7352	2.6188	7-day
Merton	1.9729	2.1888	7-day
Heston-Merton	2.8228	2.3290	1-day
GARCH-Merton	1.4819	2.9708	7-day

AAPL — RMSE(S_t)

Model	7-day	1-day	best window
GBM	0.6704	1.3769	7-day
Merton	0.7940	1.2217	7-day
Heston-Merton	2.3470	1.4030	1-day
GARCH-Merton	0.9724	1.5616	7-day

MSFT — RMSE(S_t)

Model	7-day	1-day	best window
GBM	1.3803	1.8735	7-day
Merton	1.4984	1.1616	1-day
Heston-Merton	5.7987	3.3681	1-day
GARCH-Merton	1.2807	1.6015	7-day

Optimal-stopping RMSE: LSM American call vs Black-Scholes European. 12 synthetic ATM/OTM/ITM contracts per name, expiry 2023-03-15 15:59, 500 LSM paths.
No 2023 listed quotes in the project options panel. Same contracts and seeds across models.

SPY — RMSE(LSM vs BS)

Model	7-day	1-day	best window
GBM	0.3156	0.1522	1-day
Merton	0.3062	0.1452	1-day
Heston-Merton	9.5968	4.9914	1-day
GARCH-Merton	0.4212	0.1509	1-day

AAPL — RMSE(LSM vs BS)

Model	7-day	1-day	best window
GBM	0.1406	0.0792	1-day
Merton	0.1296	0.0792	1-day
Heston-Merton	1.9479	2.2550	7-day
GARCH-Merton	0.2334	0.1087	1-day

MSFT — RMSE(LSM vs BS)

Model	7-day	1-day	best window
GBM	0.2121	0.1544	1-day
Merton	0.2083	0.1528	1-day
Heston-Merton	9.3390	2.5825	1-day
GARCH-Merton	0.5650	0.1216	1-day